

Introduction

The second leading cause of death in the U.S. is cancer, despite perpetual treatments using chemotherapy and radiation. Chemotherapy is commonly used to induce cytotoxic stress in tumor cells, often by disrupting DNA replication or cell division. Tumor deterioration is dose-dependent, which contributes to its ability to develop resistance (Bae et al., 2020). In fact, variability in cell survival is mainly attributed to the heterogeneity of RNA and resistance protein production. This is a key contributor to cancer cell plasticity among various drug concentrations. While many forms of biological resistance are assigned to mutations that are hard-coded into the genome and prompt cells to evolve built-in defense responses, tumor resistance is not always driven by irreversible mutations. Intrinsic and reversible stochasticity can cause phenotypic changes in resistance protein production and cell fate measured by drug sensitivity (Kumar et al., 2019). This phenomenon is known as stochastic tuning.

Stochastic tuning is a non-genetic mechanism that generates transcriptional noise and causes random bursts in gene expression to develop cell plasticity. These dynamics allow cells to modulate transcriptional variability and promote survival under stressful conditions. There is crucial evidence of the adaptation of this mechanism by the model yeast organism, *Saccharomyces cerevisiae*, as a defense mechanism against extreme environmental conditions. Eukaryotic cells typically have predefined regulatory gene networks that control gene expression states for specific conditions through transcriptional memory. However, this can fail when cells are exposed to unfamiliar conditions (Freddolino et al., 2018). Examples include extreme temperature, pH, and salinity. This defense mechanism of variable transcriptional noise evolved past single-celled organisms. It provides an explanation for multicellular processes like cell differentiation and embryonic development, a multicellular effort in optimizing tissue and organ patterns. Cytotoxicity is one example of a multicellular effort in stochastic tuning to maintain survival by modifying chromatin architecture and gene expression. This is understood through the upregulation of DNA synthesis genes like dihydrofolate reductase in the presence of drugs like MTX, where resistance genes are those that normally promote cell proliferation when expressed (Cowan et al., 1986).

In fact, stochastic tuning is not only limited to the gene level. It also provides an explanation for cell fate. One example of this is epithelial-mesenchymal transition (EMT), a process associated with switching between drug-sensitive and drug-resistant phenotypes respectively. This also explains drug resistance, where epithelial cells are fast-growing in organs and tissues, and mesenchymal cells are highly migratory and associated with metastasis (Kumar et al., 2019). Motivated by how upregulation of resistance genes is a product of stochastic tuning, this project develops a computational model to investigate how variability in RNA expression and protein production influences tumor resistance under cytotoxic stress.

Model

To quantify and model the role of stochastic tuning in tumor resistance against cytotoxicity, the simulation's logic follows the Gillespie algorithm. This is a continuous Monte Carlo stochastic simulation that is commonly used to model coupled reactions. The reactions in the simulation included drug-induced gene activation and inactivation, transcription and RNA degradation, translation and degradation, cell division, and drug-induced death. Since the stochasticity in cell fate is largely influenced by that at the gene level with various drug concentrations and fixed rates for cell division and death, this model simplifies stochastic tuning by omitting the cell fate switching. It tracks the stochasticity of RNA and resistance proteins in a reaction system at the cell and population level, sampling the waiting time for the

next reaction in a manner that is distributed exponentially. It updates the amount of RNA, protein, and cells depending on the weight of the reaction for each computational run. The simulation loops through each dosage to quantify their respective effects on the molecular RNA, protein, and cell population.

The simulation starts by defining the list of dosages that will affect gene expression in each cell of the tumor: 1 μM , 10 μM , 50 μM , 100 μM , 250 μM , 500 μM , 1000 μM . The simulation works with 50 cells, and an array of zeros of the same size as the number of cells is initialized for gene state (x), RNA, and protein. The simulation simultaneously tracks single-cell trajectories and population-level behavior to analyze stochasticity at the gene level while analyzing population patterns to determine tumor resistance as a whole. Single-cell behavior was tracked by assigning the variable “tracked” to one cell_id and initializing a list to track that cell’s number of RNA molecules, protein molecules, and lifetime in seconds. Additionally, there was a list for overall RNA, protein, and time to analyze population behavior. In order to make experimental snapshots, the number of RNA and proteins were sampled every five seconds. Following that, constant parameters would determine reaction rates. The two reactions that were affected by the drug concentration were gene activation and cell death.

Equation 1. Drug-induced gene activation

$$K_{on} = drug_{delta} + 0.2 * D_{current} / (k_d + D_{current})$$

Equation 2. Drug-induced death rate modeled by the Hill function

$$death = drug_{delta} * ((D_{current}^n) / (D_{current}^n + (K * prot)^n))$$

The death rate was modeled by the Hill function, a mathematical equation that is often used in pharmacology to illustrate the relationship between drug concentration and some biological effect through a sigmoidal curve (Lipan and Ferwerda, 2018). In this simulation, it specifically resembles the inhibition rate, where the resistance protein acts as a competitive inhibitor to drug-induced apoptosis. The activation rate was modeled by adding drug_delta to a ratio that determines the drug efficacy at 50% success, where 0.2 is a constant alpha value that was picked to produce optimal bursting trajectories. The parameters that went into each reaction are enlisted in Table 1.

The simulation’s core logic is driven by a while loop that terminates when the time limit is reached (50 seconds), all cells die, or a maximum population threshold is reached (10000 cells) for run time purposes. Within each iteration, the model determines the cell fate (division/death), protein production, RNA production, and gene state using a stochastic event selection process. Following the Monte Carlo logic, “r” represents a specific event based on the total rates weighted by a sampled random number. This cycles through the reaction space to determine which event happens and which cell is affected. Importantly, the Gillespie algorithm samples the waiting time between events as a random number.

The conclusions made from the model rely on major assumptions and shortcomings that do not reflect the real world. For example, cancer stages were omitted from this model as a generalization of the relationship between stochastic tuning and disease persistence specifically focused on drug-induced transcription. Moreover, there is a limited number of RNA and proteins in the cell to accommodate for expensive run time. Additionally, in terms of the bigger picture with treatment cycles, drug administration and rest periods were not considered.

To validate and visualize this model, the simulation generates time-series plots of RNA and protein at the single-cell and population level, a semilog graph that evaluates the relationship between drug dosage and tumor resistance via cell survival, and histograms of RNA and protein distributions in

comparison to a Poisson distribution curve to analyze statistical properties such as variance and Fano factor.

Table 1. Summary of all model parameters, their values, and biological interpretations.

Parameter	Interpretation	Equations	Simulation Values
division	division constant		0.02
k_on	Activation constant	$(\text{drug_delta} + 0.2 \cdot \text{D_current}) / (k_d + \text{D_current})$	
k_off	Inactivation constant		100
ktransc	Transcription constant		100
ktransl	Translation constant		70
mrna_decay	mRNA degradation constant		0.1
prot_degrad	Protein degradation constant		0.5
drug_delta	Maximum death rate drug can achieve		0.01-1
K	Protein level to beat the effects of the drug		0.001
D_current	Current dosage		1, 10, 50, 100, 250, 500, 1000
death	Drug-dependent death rate	$\text{drug_delta} \cdot ((\text{D_current}^n) / (\text{D_current}^n + (K \cdot \text{prot})^n))$	
n	Hill coefficient resembling the strength of the dose-reponse		1, 1.5, 2
x	Array of gene states		
RNA	Array of mRNA		
prot	Array of proteins		
active	Activation rate	$k_{\text{on}} \cdot (1-x)$	
off	Inactivation rate	$k_{\text{off}} \cdot x$	
transc	Transcription rate	$k_{\text{transc}} \cdot x$	
transl	Translation rate	$k_{\text{transl}} \cdot \text{RNA}$	
mrna_death	mRNA degradation rate	$\text{mrna_decay} \cdot \text{RNA}$	
prot_death	Protein degradation rate	$\text{prot_degrad} \cdot \text{prot}$	
div	Division rate	$\text{division} \cdot \text{ones}(1, N)$	
N	Number of cells		50-100
tmax	Maximum lifetime of a cell		50
kd	induction midpoint dose where 50% of cells are killed		

Results

To analyze the stochasticity in RNA production, protein production, and final population, the parameters were set to an inactivation rate of 100, $K = 0.001$, transcription rate of 100, translation rate of 70, division rate of 0.02, mRNA decay rate of 0.1, a protein decay rate of 0.5, and $\text{drug_delta} = 0.1$, as Table 1 details. These parameters were purposely chosen to produce visible bursting patterns. Moreover, results represent five stochastic simulations run for 50 seconds each across drug concentrations.

Figure 1. Single-cell and population RNA and Protein Time Series at $n=1$

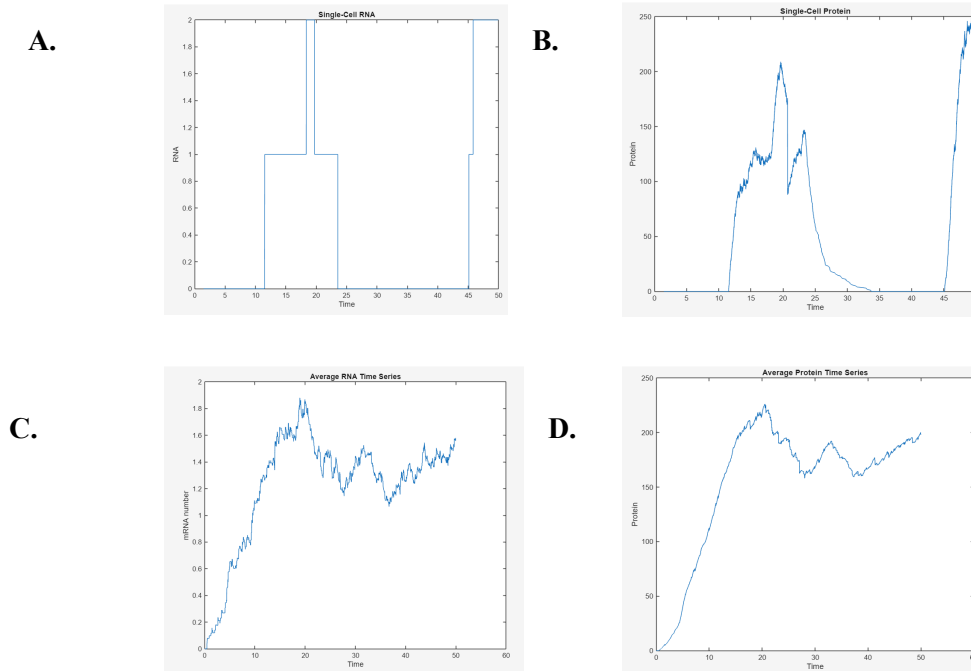


Figure 1. Plots A and B show representative single-cell trajectories for RNA and protein expression, respectively. Plots C and D show average trajectories of RNA and proteins per cell. Fluctuations in RNA abundance propagate to protein expression, generating substantial heterogeneity over time. Across multiple runs, the level of bursting follows various patterns, but the overall trend in these trajectories show noise generating variable RNA and protein numbers.

Figure 2. Dose Response under varying Hill coefficients (n=1, n=1.5, and n=2)

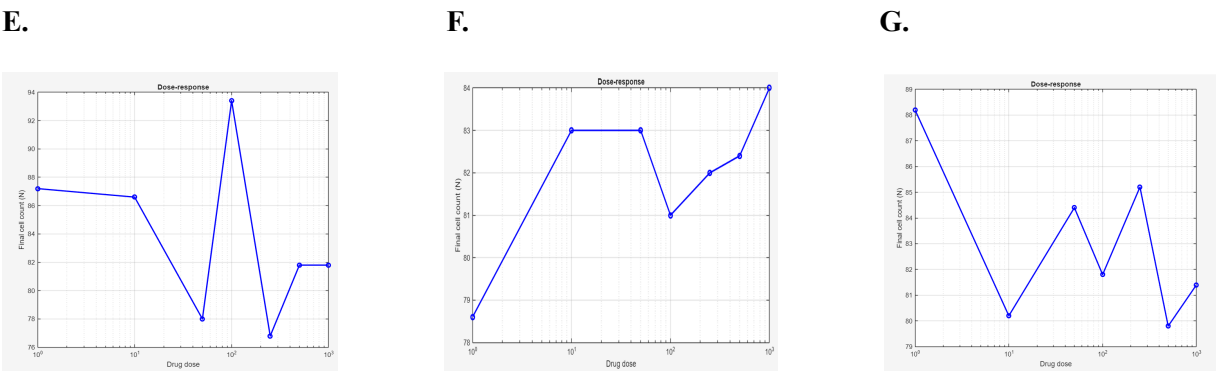


Figure 2. Each figure shows the surviving population for each drug concentration from five trials in the simulation. Survival responses vary substantially across different hill coefficients, indicating sensitivity of population dynamics to transcriptional regulation and stochastic fluctuations in response to various drug concentrations.

Figure 3. RNA and protein distribution probabilities at the population level at n=1 and n=2

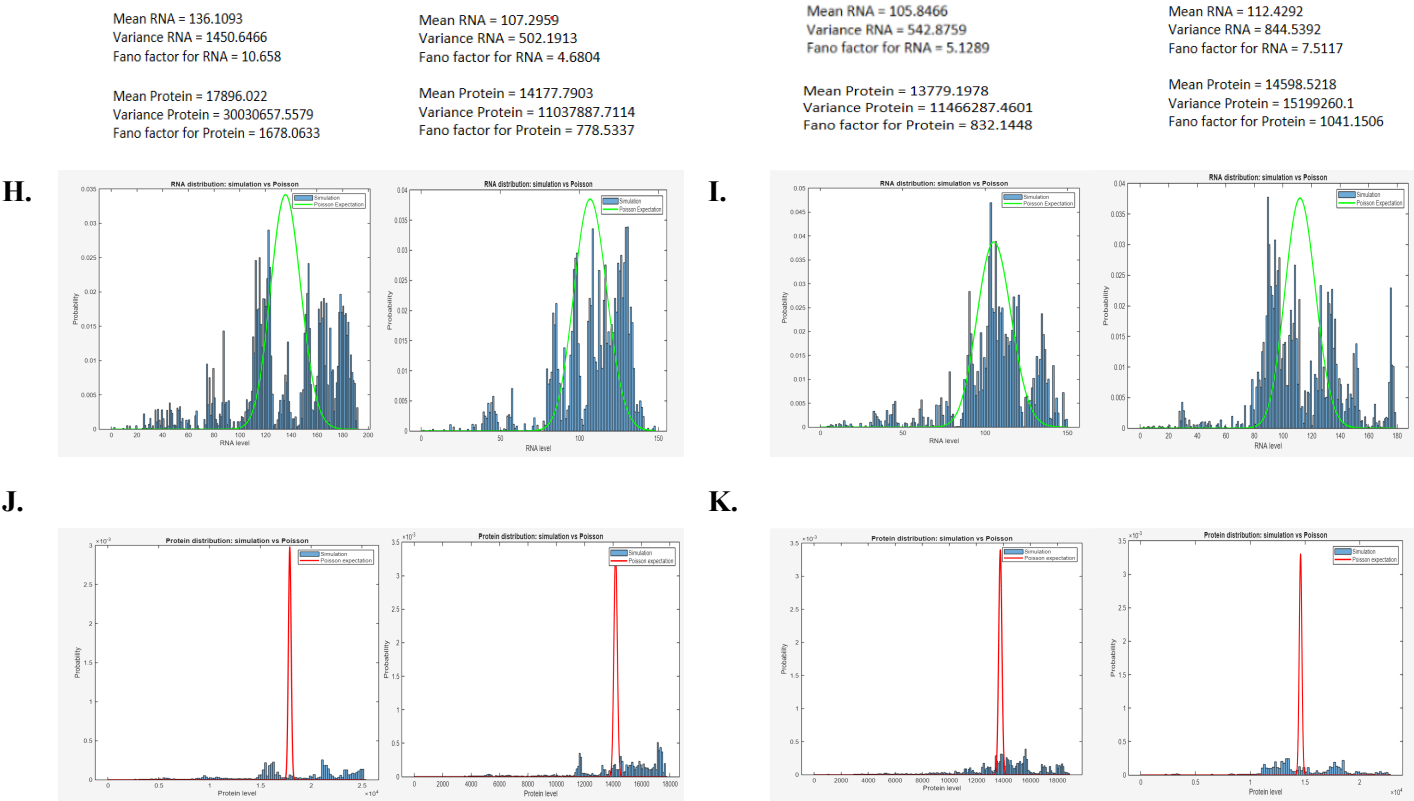


Figure 3. Plot H shows some bimodality in RNA distribution at $n=1$ in the first run. It is overdispersed and peaks at two extremes of RNA distribution (e.g. 100 and 200 molecules). However, this is not maintained across runs. Instead, substantial overdispersion relative to the Poisson distribution persists across all the plots in Figure 3, as the fano factor remains much greater than 1. Moreover, there is no indication that a changing hill coefficient correlates to a clear pattern in RNA and protein distribution.

Discussion

The plots in Figure 1 support variability in RNA and protein expression per cell through burst-like dynamics in single-cell trajectories. For example, these fluctuations arise from stochastic switching between gene activation and inactivation states, particularly under conditions where inactivation rates exceed activation rates, as was the case in the above results. This was especially true because gene activation was limited by the maximum death rate achieved by the drug. The single-cell results emphasize that gene expression exhibits extended periods of quiescence followed by short bursts of high activity consistent with transcriptional bursting behavior. The average population results emphasize the pattern of increasing molecules and fluctuation at a level that optimizes cell survival.

In fact, since parameters were set to optimize survival with long inactivation times, it is worthy to note that the cells do not completely die by the end of the simulation in Figure 2. While unpredictable, the dose-response curves show how the unpredictability facilitates cell survival despite various cytotoxic effects. Contrary to the expectations that fewer cells survive as the hill coefficient increases the affinity between drug and protein, the survival patterns remain stochastic, reflecting the increasing likelihood of the population's ability to rebound and for cells to continue dividing while their RNA and protein contents also fluctuate. While there is an increasing trajectory towards the end of plot F near 1000 μm , this is not always consistent across runs and requires further parametrization manipulation.

At the population level, Figure 3 shows that RNA and protein distributions exhibit strong overdispersion relative to the Poisson distribution. This is supported by Fano factors much greater than one. This suggests that intrinsic noise dominates expression variability rather than an underlying Poisson transcription process. While the distributions show heavy tails, they are sometimes bimodal when $n=1$ and mostly overdispersed when $n=2$, which indicates a broad spectrum of expression states rather than discrete switching to escape death. The large fano factors and variance reflect super-Poissonian behavior commonly associated with transcriptional bursting and other stochastic biological processes (Kumar et al., 2019). This suggests that the tumor transitions between heterogeneous expression states, increasing the likelihood that subsets of cells transiently tolerate diverse cytotoxic stress. It matches the main goals of stochastic tuning as an adaptation mechanism in response to unexpected conditions.

Despite simplifying assumptions such as excluding EMT dynamics, maintaining a finite population size, ignoring drug treatment schedules, limiting trials, and choosing to analyze one set of parameters for bursting efficiency, the model captures key features of non-genetic heterogeneity observed in tumor systems. It highlights how stochastic gene expression has the power to generate non-genetic variability as a drug response.

Therefore, the results are consistent with the speculation that stochastic tuning contributes to phenotypic heterogeneity and transient resistance. This is key information in current oncology and biotechnology settings, where non-genetic variability can be harnessed to shape treatment outcomes that target chemoresistance instead of specific stages of the disease.

Future directions in this simulation would be to incorporate the stochasticity in EMT and larger molecule numbers. Some literature also discusses modeling the drug-protein dynamic using a sigmoid

function, so it would be helpful to compare the change in stochastic gene expression under the influence of the different mathematical models (Farquhar et al., 2019). Finally, while the simulation suggests that stochastic molecular dynamics contribute to tumor resistance in an unpredictable manner, this is mainly because extreme parametrization past the current optimal settings must be manipulated to determine if this stochasticity can be controlled. This can be achieved through more frequent trials and various levels of drug-protein affinity.

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